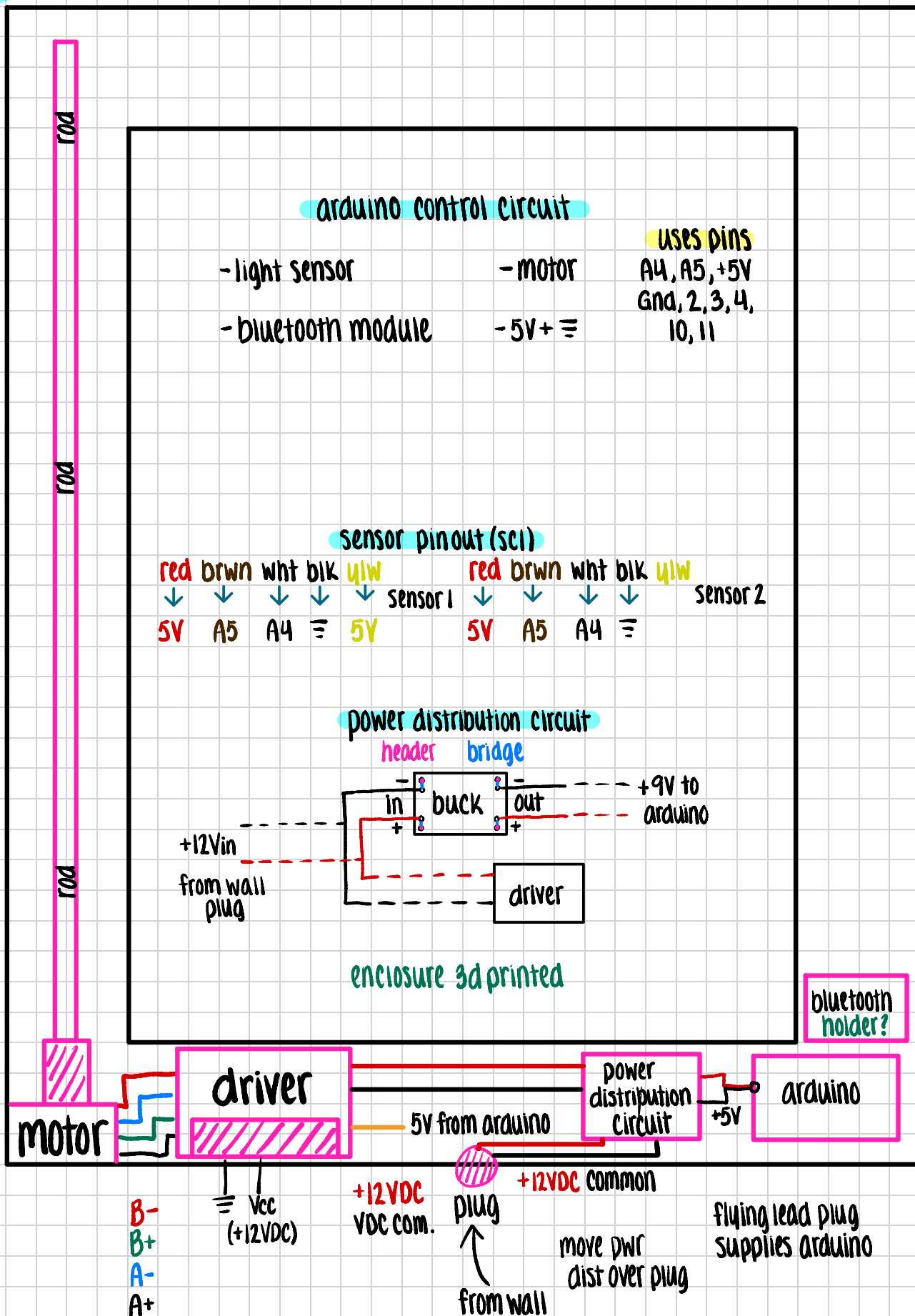


frame elect. layout

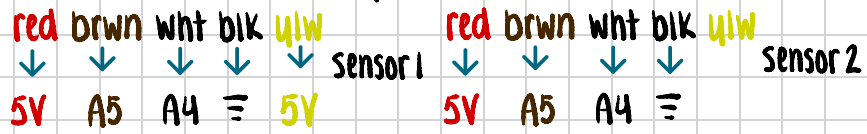
maybe 3D print: lux bracket
pwr dist bracket
arduino brackets

wip 2/25/26

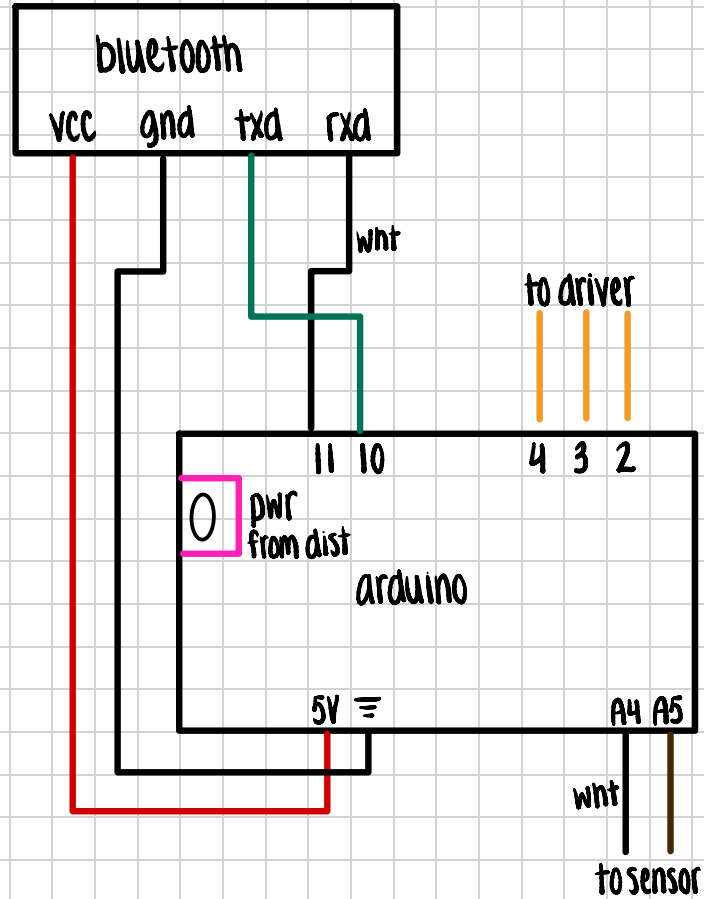
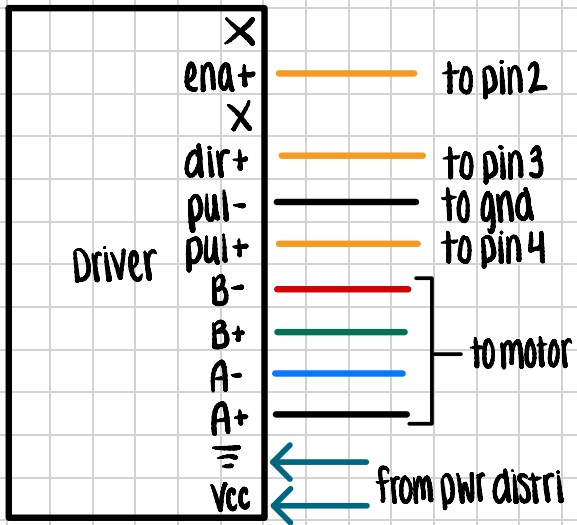


needs secured
pwr dist circ
arduino
bluetooth
lux sensors

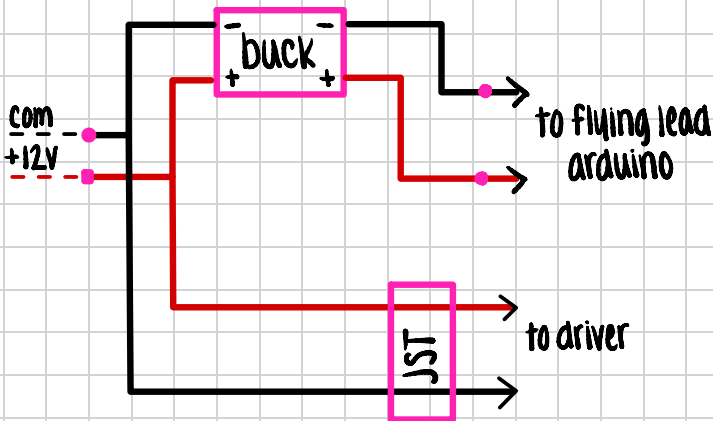
sensor pinout (sel)



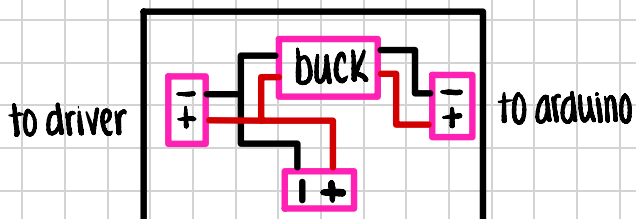
Driver & arduino pinout

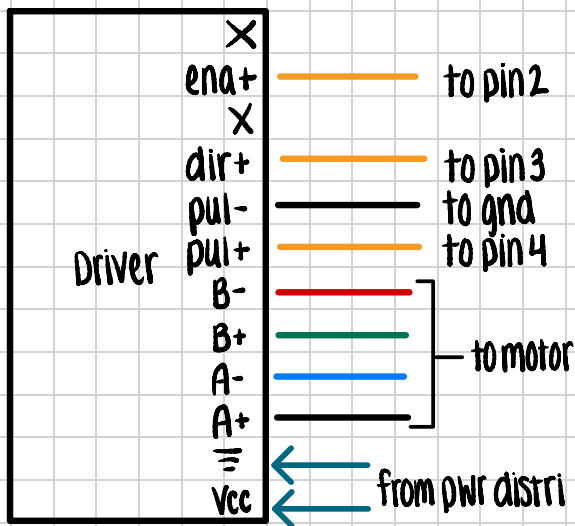
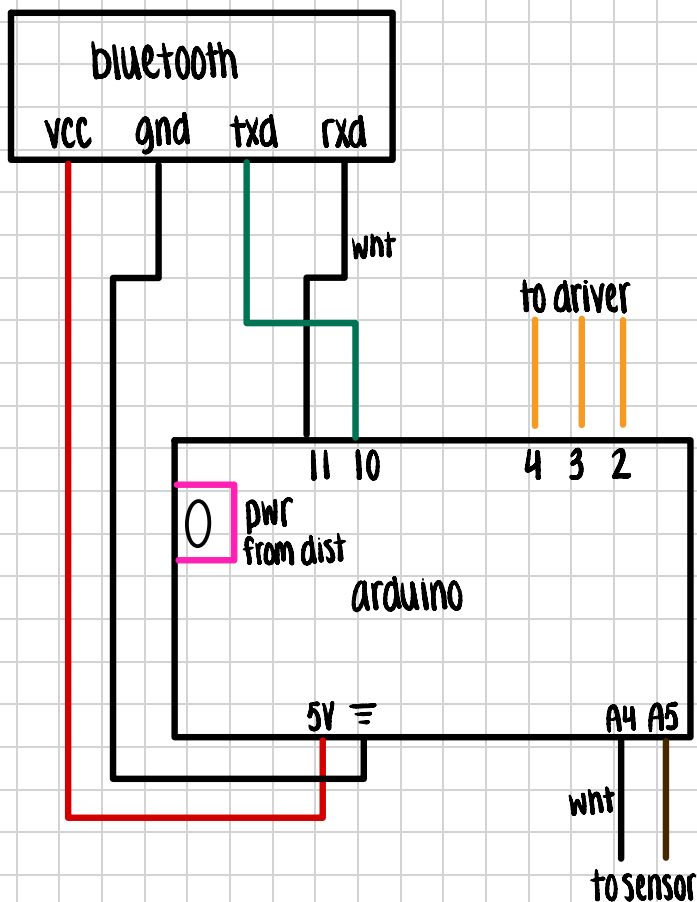


pwr distribution



control circuit





Arduino Connections:
 Arduino +5V -- VCC node
 Arduino GND -- GND node
 Arduino A4 -- A4 node
 Arduino A5 -- A5 node

BH1750FVI #1 Connections:
 VCC -- VCC node
 SCL -- A5 node
 DAT -- A4 node
 GND -- GND node

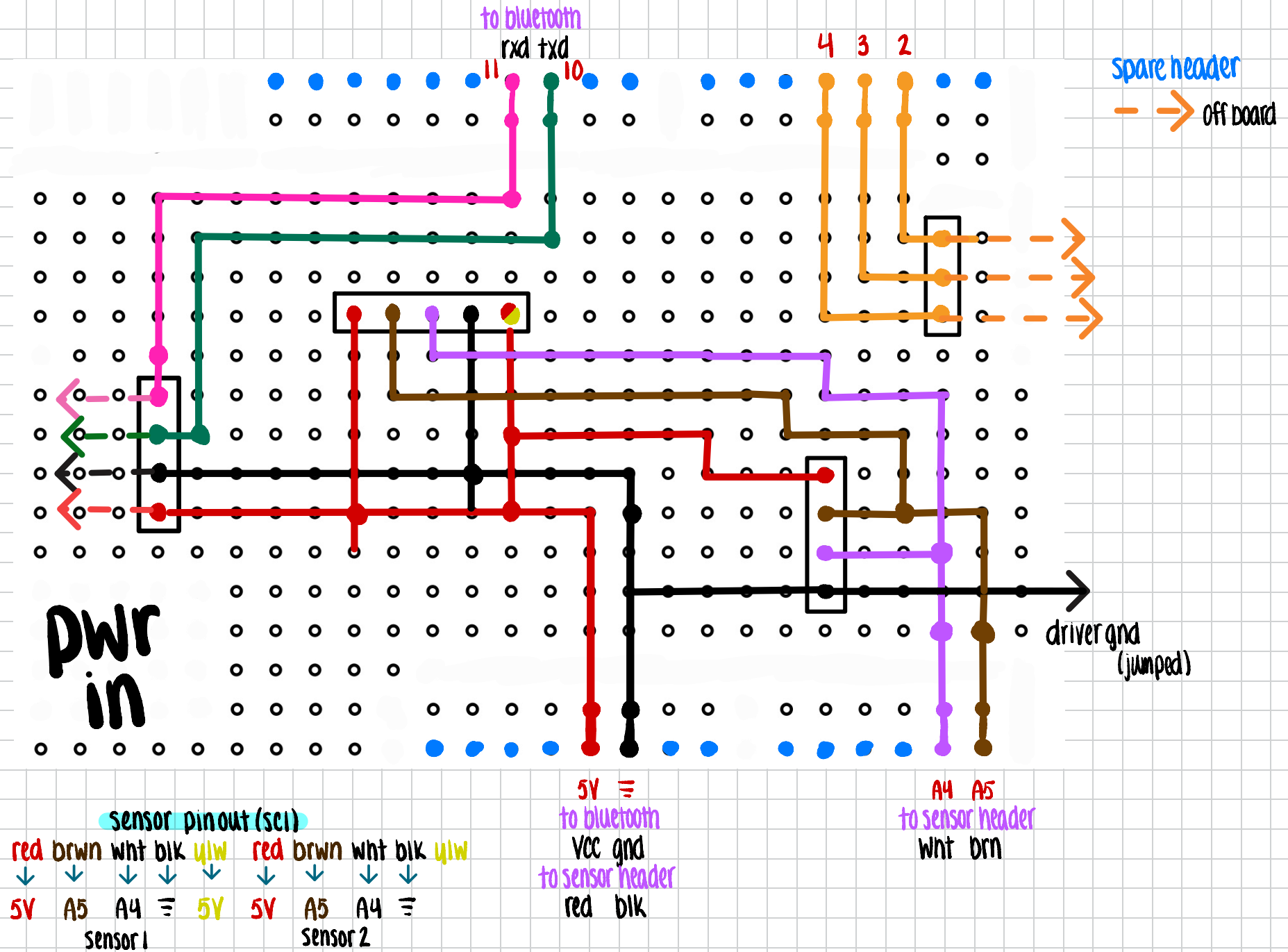
BH1750FVI #2 Connections:
 VCC -- VCC node
 SCL -- A5 node
 DAT -- A4 node
 GND -- GND node
 ADDR -- VCC node

Motor Driver Setup:
 Set Current to 1.0A
 Set Pulse/rev to 6400
 1 (S1) -> OFF
 2 (S2) -> OFF
 3 (S3) -> OFF
 4 (S4) -> ON
 5 (S5) -> OFF
 6 (S6) -> ON

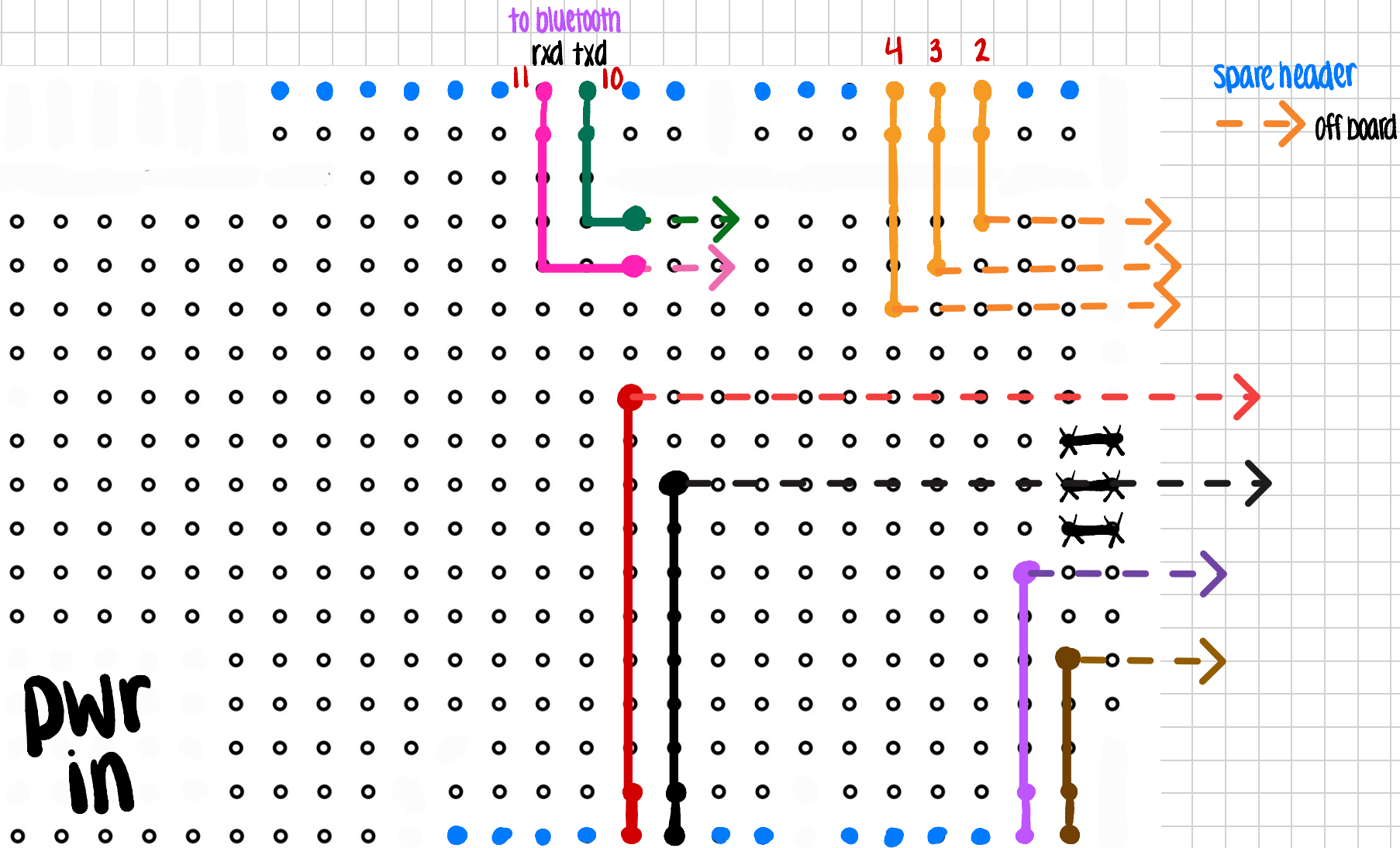
Motor Driver/Microcontroller Connections:
 TB66000 (ENA-) -- TB66000 (DIR-)
 TB66000 (ENA+) -- Arduino 4
 TB66000 (DIR-) -- TB66000 (PUL-)
 TB66000 (DIR+) -- Arduino 2
 TB66000 (PUL-) -- Arduino GND
 TB66000 (PUL+) -- Arduino 3

Motor/Motor Driver Connections:
 Motor Red Wire (B+) -- TB66000 (B-)
 Motor Green Wire (B+) -- TB66000 (B+)
 Motor Blue Wire (A-) -- TB66000 (A-)
 Motor Black Wire (A+) -- TB66000 (A+)
 Power Supply (GND) -- TB66000 (GND)
 Power Supply (+12V) -- TB66000 (VCC)

HM-10 Bluetooth Module:
 RXD -- Arduino 11
 TXD -- Arduino 10
 GND -- GND node
 VCC -- VCC node



new
plug
method



PWR
in

sensor pinout (sci)

red	brwn	wht	blk	ylw	red	brwn	wht	blk	ylw
5V	A5	A4	≡	5V	5V	A5	A4	≡	
Sensor 1					Sensor 2				

plug
method

driver

bluetooth

sensor

spare header

— → off board

sensor pinout (sc1)

red	brwn	wht	blk	ylw	red	brwn	wht	blk	ylw
↓	↓	↓	↓	↓	↓	↓	↓	↓	
5V	A5	A4	≡	5V	5V	A5	A4	≡	
	Sensor 1					Sensor 2			

